BBM 1104 · The Smart Ledger System — Completed Storyboard

1 · Cover & Metadata

FIELD I SUMMARY	
Course/topic	BBM 1104 - Principles of Accounting | Double-entry bookkeeping: debit and credit rules, ledger posting, the trial balance, and error correction
Simulation title	The Smart Ledger System
SME / Instructional Designer	Obiadi Chinaza
Designer / Builders	Instructional Design & Simulation Development Team
Date/version	7 October 2026 | Storyboard draft v1.0
Target learners	Undergraduate students; basic accounting vocabulary assumed (assets, liabilities, capital, income, expenses). No prior experience with accounting software is required.
Estimated duration	20-25 minutes across five scenes, excluding the facilitator debrief
Setting	Zuri Home Supplies, a fictional homeware shop in Nairobi, Kenya, piloting an automated bookkeeping prototype called Smart Ledger. The learner works as a junior accounts systems assistant.
Learning outcomes	Learners will be able to: (1) apply debit and credit rules by account type to common business transactions; (2) place ledger balances in a trial balance and identify the size of a difference; (3) trace a difference to its cause using a suspense account; (4) correct one-sided, wrong-amount, reversed and wrong-account errors; (5) test a posting system and explain what a balanced trial balance can and cannot show.

2 · Simulation Type Selector

Tick and combine as needed (see Instructions Guide, Section 3, for full descriptions):

☒ Dialogue/branching
☒ Evidence-gathering / environment-emergent exploration
☒ Instrumented testing
☒ Procedural sequencing
☐ Token/resource allocation
☐ Annotation builder
☒ Others: scenario evaluation/reflection (Scene 5 optional reflection)

Tone, voice & reading level

Professional, supportive and practical Kenyan business context; British English and accessible undergraduate language; short, natural spoken sentences. Accounting terms such as debit, credit, ledger, posting, trial balance, suspense account, and transposition are introduced through realistic transactions and reinforced through T-account traces and feedback. Use the account names provided in the activity to avoid ambiguity between alternative accounting conventions.

4 · The Premise

The setting/world

Zuri Home Supplies is a growing homeware shop in Nairobi. Owner Grace Achieng sells kitchenware and household goods for cash and on credit, buys stock from suppliers, pays rent and delivery costs, and sometimes takes money out for personal use. To keep up, the shop has started using Smart Ledger, a prototype system that posts each transaction to ledger accounts automatically.

The learner's role

The learner is a junior accounts systems assistant assigned to review, repair, and test the system's posting logic before Grace relies on it for month-end.

The problem/situation

After the first month, the trial balance does not balance, and the system has parked the difference in a suspense account. Some transactions are posted with the debit and credit sides swapped or to the wrong account, and the system cannot explain its own results.

The goal/objective

Set accurate posting rules from the account-type rules, trace and correct the errors in the ledger, test that the fixes work, and explain how the system reaches its results and what a balanced trial balance can and cannot prove.

The posting rule sheet (fictional; fixed for the simulation)

POSTING RULE SHEET	
Assets: Cash, Bank, Inventory, Equipment, Trade Receivables	Increase with a debit; decrease with a credit.
Expenses: Rent Expense, Delivery Expense	Increase with a debit; decrease with a credit.
Drawings	Increases with a debit when the owner takes money or goods for personal use.
Liabilities: Trade Payables	Increase with a credit; decrease with a debit.
Capital	Increases with a credit when the owner invests.

POSTING RULE SHEET	
Income: Sales	Increases with a credit.
Every posting	Total debits must equal total credits. Smart Ledger supports bookkeeping checks only; an accountant reviews the accounts.
Opening balances (1 June)	Cash 20,000 Dr; Bank 150,000 Dr; Inventory 80,000 Dr; Equipment 100,000 Dr; Capital 350,000 Cr. Totals: 350,000 on each side.

5 · State & Scoring Framework

ACCURACY / STATE STRANDS	
Strand 1: Posting-Rule Accuracy (PR) – 30%	Ability to choose the correct debit and credit accounts for each transaction type using the account names provided. Updated in Scene 2.
Strand 2: Trial Balance & Ledger Skills (LT) – 15%	Ability to place ledger balances in the correct trial balance column and identify the size and direction of a difference. Updated in Scene 3.
Strand 3: Error Detection & Correction (ED) – 25%	Ability to identify error types and choose correcting entries using a suspense account. Updated in Scene 3.
Strand 4: Testing & Debugging (TD) – 25%	Ability to design test cases, predict results before running, and revise the faulty rule or routine. Updated in Scene 2 (10% of the strand, the audit approach) and Scene 4 (90%).
Strand 5: Reasoning (RS) – 5%	Ability to explain the posting rules, the trial balance trace, and what a balanced trial balance cannot show. Updated in Scene 4 (the handover rationale).

Scoring model

Final score = (Posting-Rule Accuracy x 30%) + (Trial Balance & Ledger Skills x 15%) + (Error Detection & Correction x 25%) + (Testing & Debugging x 25%) + (Reasoning x 5%). All tasks, learning content, and scored decisions take place in Scenes 2-4. Scenes 1 and 5 award no points. Scores come only from the strand percentages shown here.

Running log (if applicable)

Smart Ledger Test Log. The log stores rule-mapping choices, the audit action, trial balance column choices, error types and correcting entries, test predictions, actual results, revisions, hints, and the handover rationale. It is used for the test report in Scene 4 and for feedback in Scene 5; the optional reflection is saved but not scored.

6 · Structure & Flow

Duplicate the node table below once per scene. Keep field order fixed: Situation / Interaction / Response indicator / Teaching content / State change / Goes to.

STANDARD MIVA SHELL

OPENING	OPTIONAL INTRO SEQUENCE	TITLE / LANDING PAGE / Student Name	SIMULATION SCENES 1–5

The opening shell does not count toward the five primary simulation scenes. Scene 1 is a non-interactive opening sequence and title card (delivered through the shell's title/landing page). Scenes 2-4 contain all tasks and learning content, each with feedback and state updates. Scene 5 is the debrief, feedback, and closing scene. Scores come only from the strand percentages in Section 5.

Scene [1] — [Opening Sequence / Title Card – Welcome to Zuri Home Supplies]

NODE [1] — [Opening Sequence / Title Card – Welcome to Zuri Home Supplies]	
Situation	**Visuals:** Fade in from a Nairobi street in the morning and enter Zuri Home Supplies in a short cinematic sequence. The viewer sees shelves of kitchenware, a till, supplier invoices, a notebook of handwritten sales, a mobile-money phone, and a laptop showing a dashboard labelled "Smart Ledger – Pilot". Grace Achieng, the owner, studies the laptop while a customer pays. The sequence settles on the title card: "The Smart Ledger System", with the learner's name and the role "Junior Accounts Systems Assistant". **Audio:** door chime, soft shop ambience, till and keyboard sounds, phone notification, light background music. A short title card sting plays as the title appears. **Scenario text on screen:** "Zuri Home Supplies has started using Smart Ledger to record its transactions automatically. Grace wants to know whether she can trust it." **Grace:** "The system records everything for me now. But the trial balance does not balance, and I cannot tell whether the postings are right." **Narrator:** "You are joining Grace as a junior accounts systems assistant. Your task is to set the posting rules, trace the errors in the ledger, then test the fixes and hand over. You will be assessed on posting rules, ledger and trial balance skills, error correction, testing, and your explanation. Speed is not scored. All figures are fictional."
Interaction	**Non-interactive opening sequence and title card.** The learner watches or listens to the sequence (about 60-75 seconds). No hotspots, transactions, calculations, or choices appear in this scene.

NODE [1] — [Opening Sequence / Title Card – Welcome to Zuri Home Supplies]	
	Standard landing-page controls only: the learner's name from the landing page appears on the title card; subtitle and audio settings; and a "Start" button that opens Scene 2. These are navigation controls, not learning interactions.
Response/indicator	The title card fades in at the end of the sequence, and "Start" becomes active. No correctness feedback, indicators, or scores are shown in this scene.
Teaching content	**Knowledge delivered:** None. No instructional content is delivered, and nothing is assessed in this scene. **Purpose:** Set the context and tone, introduce Zuri Home Supplies and Grace, state the learner's role and the plan, and name the five assessment areas. All learning takes place in Scenes 2-4.
State change	Logs learner_id, simulation_id, learner_role = junior_accounts_systems_assistant, scene_1_complete = true, and a start timestamp. No strand score changes; all strands start at 0.
Goes to	→ Scene 2 - Set the Posting Rules

Scene [2] — [Set the Posting Rules]

NODE [2] — [Set the Posting Rules]	
Situation	**Visuals:** The learner opens the Smart Ledger workspace. The posting rule sheet sits beside five sample postings produced by the draft system, shown as debit and credit lines, with the account list alongside. A "Trial Balance" panel is greyed out until Scene 3. **Audio:** keyboard clicks, soft interface tones, and a light investigative music bed. **Grace:** "Some of these postings look wrong to me, but I cannot tell why. What should the system be doing?" **Narrator:** "Read the rules first. Then check what the system actually posted."
Interaction	**Part A – Audit:** Open the rule sheet and read each account-type rule; tap each of the five sample postings to see what the draft system posted. **Part B – First debugging action:** Choose (A) change the report headings only; (B) compare each posted entry against the account-type rules; or (C) hide the accounts that look wrong. **Part C – Configure the rules:** For each of the six transaction types, choose the debit account and the credit account from the account list

NODE [2] — [Set the Posting Rules]	
	then submit the rule to place it in the posting rule table. No typing is required.
Response/indicator	**Spot on:** Selecting B opens a rule-to-posting checklist and highlights the three mismatches (S-02, S-03, S-04). Correct rule choices show the entry as a T-account preview with the debit and credit sides labelled and a one-line reason tied to the account type. **Watch out:** Choosing A or C triggers feedback that changing headings or hiding accounts does not show whether the postings are right (retry allowed). A reversed entry is named as "sides swapped"; a wrong account names the account type involved; each allows retry, with a hint after a second attempt. **Note:** None of the rule errors in this scene changes the total of debits and credits, so a trial balance would not reveal them. Scene 3 shows errors that do unbalance the trial balance, and Scene 4 tests the errors that do not.
Teaching content	Every transaction has two equal effects, a debit and a credit. The side that increases an account depends on its type: assets and expenses increase with a debit; liabilities, capital, and income increase with a credit; drawings increase with a debit. When both sides are recorded, the accounting equation stays in balance. A system applies its rules the same way every time, so one wrong rule repeats the same error on every transaction of that type.
State change	Update Testing & Debugging (10% of the strand: choosing rule-by-rule comparison) and Posting-Rule Accuracy (100% of the strand). Record audit_completed, first_action_choice, rule_mapping, retries, and hints_used.
Goes to	→ Scene 3 - Why the Trial Balance Fails

Scene 2 developer specification (Part A–B, audit)

SCENE 2 DEVELOPER SPECIFICATION (PART A–B, AUDIT)	
S-01 · Cash sale KSh 20,000	Expected: Dr Cash / Cr Sales. Draft posted: Dr Cash / Cr Sales. Finding: correct.
S-02 · Rent KSh 25,000 paid by bank	Expected: Dr Rent Expense / Cr Bank. Draft posted: Dr Bank / Cr Rent Expense. Finding: sides reversed.
S-03 · Customer pays KSh 18,000 into the	Expected: Dr Bank / Cr Trade Receivables. Draft posted: Dr Bank / Cr Sales. Finding: wrong account; sales are counted twice, and the customer's debt is never reduced.

SCENE 2 DEVELOPER SPECIFICATION (PART A–B, AUDIT)	
bank (owed from an earlier credit sale)	
S-04 · Owner takes KSh 5,000 cash for personal use.	Expected: Dr Drawings / Cr Cash. Draft posted: Dr Delivery Expense / Cr Cash. Finding: Drawings treated as a business expense.
S-05 · Stock bought on credit KSh 40,000	Expected: Dr Inventory / Cr Trade Payables. Draft posted: Dr Inventory / Cr Trade Payables. Finding: correct.
Interface	Idle ~7s: Narrator repeats the prompt. The learner must open all five postings before the first-action buttons unlock.

Scene 2 developer specification (Part C: configure the rules)

SCENE 2 DEVELOPER SPECIFICATION (PART C, CONFIGURE THE RULES)	
1 · owner invests KSh 50,000 in cash.	Dr Cash / Cr Capital
2 · Cash sale	Dr Cash / Cr Sales
3 · Stock bought on credit	Dr Inventory / Cr Trade Payables
4 · Rent paid by bank	Dr Rent Expense / Cr Bank
5 · Customer pays what they owe	Dr Bank / Cr Trade Receivables
6 · Owner takes cash for personal use	Dr Drawings / Cr Cash
Options and feedback	Account list shown: Cash, Bank, Inventory, Equipment, Trade Receivables, Trade Payables, Capital, Drawings, Sales, Rent Expense, Delivery Expense. Feedback types: sides reversed; wrong account (names the account type involved); wrong class (an expense used for drawings). Hints after a second attempt or ~8s idle; hints are recorded, not penalised. Screen readers announce “Dr” as “debit” and “Cr” as “credit.”

Scene [3] — [Why the Trial Balance Fails]

NODE [3] — [Why the Trial Balance Fails]	
Situation	**Visuals:** Smart Ledger has now posted the month's eight transactions. Ledger accounts (T-accounts) are open for Cash, Bank, Inventory, Equipment, Trade Receivables, Trade Payables, Capital, Sales, Drawings, Rent Expense, and Delivery Expense. A Trial Balance panel shows a red "Difference" badge, and a suspense account holds a debit balance of KSh 14,100. **Audio:** calculator taps, a soft alert tone when a mismatch is selected, then a low investigative music bed. **Grace:** "The system will not let me close the month. It has parked the difference in a suspense account, but nobody knows what it is." **Narrator:** "The corrected rules apply to new postings. Two faults from the first run are still in the ledger."
Interaction	**Part A – Extract the trial balance:** Sort the eleven ledger balances into the Debit or Credit column. The system then shows both totals; select the size of the difference and which column is short. **Part B – Trace the errors:** Open the three flagged accounts (Trade Payables, Delivery Expense, and Cash) and compare each with the transaction it came from. For each error, select the type: one-sided posting, transposition, reversed sides, or wrong account. **Part C – Correct and clear the suspense account:** For each error, choose the correcting entry from the options. The suspense balance updates after each correction until it reaches nil.
Response/indicator	**Spot on:** Correct column choices lock the balance into place and the totals update; correct error types open a short worked trace; correct correcting entries move the suspense balance towards nil, and the trial balance shows "Agrees." **Watch out:** a wrong column returns the card with a hint about normal balances. A wrong error type shows both sides of the entry for comparison. A wrong correcting entry shows its effect on the suspense balance (for example, "Suspense is now Dr 15,000; the first error is not yet cleared"). Partial completion leaves the suspense balance showing what remains.
Teaching content	A trial balance lists every ledger balance; if total debits and credits differ, an error exists. A suspense account holds an unexplained difference temporarily, so each error must be traced and corrected, not hidden. A one-sided posting leaves one leg missing; a transposition swaps digits, and the difference divides exactly by nine. Reversed sides and

NODE [3] — [Why the Trial Balance Fails]	
	wrong-account errors do not change the totals, so a trial balance cannot detect them.
State change	Update Trial Balance & Ledger Skills (100% of the strand) and Error Detection & Correction (100% of the strand). Record trial_balance_columns, difference_identified, error_types_selected, correcting_entries, suspense_balance_after_each_step, retries, and hints_used. Flag: ledger_balanced = true.
Goes to	→ Scene 4 - Test, Debug and Hand Over

Scene 3 developer specification

SCENE 3 DEVELOPER SPECIFICATION	
Month's transactions (June)	1) Cash sales 60,000. 2) Stock bought on credit 40,000. 3) Rent paid by bank 25,000. 4) Credit sale 30,000. 5) Customer pays into bank 18,000. 6) Supplier paid from bank 15,000. 7) Owner takes cash 5,000. 8) Delivery paid in cash 4,500.
Balances from the faulty ledger (Part A)	Debit: Cash 70,500; Bank 128,000; Inventory 120,000; Equipment 100,000; Trade Receivables 12,000; Drawings 5,000; Rent Expense 25,000; Delivery Expense 5,400 = 465,900. Credit: Capital 350,000; Sales 90,000; Trade Payables 40,000 = 480,000. Difference 14,100 (debits short). The system has opened Suspense at Dr 14,100 to hold it.
Error E1 · one-sided posting	Supplier paid KSh 15,000 from bank: Cr Bank was posted, but Dr Trade Payables was not. Trade Payables shows KSh 40,000 Cr instead of KSh 25,000.
Error E2 · transposition	Delivery costs KSh 4,500 paid in cash: Cr Cash 4,500 was posted, but Dr Delivery Expense was keyed as 5,400. The difference of 900 is divisible by 9.
Error types offered (Part B)	One-sided posting; transposition; reversed sides; wrong account. E1 = one-sided posting. E2 = transposition.
Correcting entries (Part C)	E1: Dr Trade Payables 15,000 / Cr Suspense 15,000. E2: Dr Suspense 900 / Cr Delivery Expense 900. Suspense moves from Dr 14,100 to Cr 900 after E1 only, to nil after both. Correcting E2 only leaves Dr 15,000.
Distractor corrections	Crediting Bank again for 15,000 (double counts the payment); writing the whole 14,100 off to Delivery Expense; posting the difference to one account without tracing it.

SCENE 3 DEVELOPER SPECIFICATION	
Corrected trial balance	Debit total 465,000 = credit total 465,000. Trade Payables 25,000 Cr; Delivery Expense 4,500 Dr; Suspense nil.

Scene [4] — [Test, Debug and Hand Over]

NODE [4] — [Test, Debug and Hand Over]	
Situation	**Visuals:** The corrected ledger closes, and the system opens a test console beside the rule table. The console lists ten test transactions in four categories (normal, rule check, routine fault, and balanced but wrong), with columns for expected entry, actual entry, and trial balance status. When testing is complete, Grace asks for a brief explanation of how the system reaches its postings and whether a balanced trial balance can be trusted. **Audio:** keyboard clicks, a soft pass tone and a soft alert tone, light background music. **Grace:** "The books balance now. How do I know the system will not make the same mistakes next month?"
Interaction	**Part A – Test:** Choose test cases (at least one from each of the four categories), predict the debit and credit accounts before running, run the system, and compare the actual result with the expected one. When a mismatch appears, select the rule or routine to revise. **Part B – Regression:** Run all ten cases against the finished system and confirm the test report. **Part C – Handover:** Choose the final summary and give a short rationale: name the debit and credit rule by account type, explain how a trial balance difference is traced, and state what a balanced trial balance cannot show.
Response/indicator	**Watch out:** A routine fault shows the missing leg, wrong figure, or wrong account in a side-by-side comparison. In the balanced-but-wrong case the console reports "Balanced," and feedback asks the learner to compare the entry with the expected account. If the explanation overclaims ("the trial balance balances, so the books are right"), feedback asks the learner to restate what a trial balance can and cannot show. A failing case routes back to the relevant scene panel. **Spot on:** Correct revisions update the displayed test report; complete test evidence and a clear explanation produce a handover-ready report.
Teaching content	Testing should include typical transactions, the rules already corrected, the faults found in the ledger, and cases where the trial balance would still balance. Correcting an error in the ledger is not the same as fixing the routine that caused it, so the cause must be tested again. A result is

NODE [4] — [Test, Debug and Hand Over]	
	correct only when the entry follows the posting rules, not merely because debits equal credits. The system implements only the fictional rules provided and supports bookkeeping checks only. This scene draws together the rules (Scene 2) and the error tracing (Scene 3).
State change	Update Testing & Debugging (90% of the strand) and Reasoning (100% of the strand). Record cases_chosen, predictions, actual_results, revisions, hints_used, final_summary_choice, rationale, and test_evidence. Flag: handover_complete = true.
Goes to	→ Scene 5, Debrief, Feedback & Closing, after all ten cases pass and the handover is submitted.

Scene 4 developer specification (Parts A–B, testing)

SCENE 4 DEVELOPER SPECIFICATION (PARTS A–B, TESTING)	
T1 · Normal	Cash sale KSh 12,000. Expected: Dr Cash / Cr Sales. Draft: matches.
T2 · Normal	Credit sale KSh 30,000. Expected: Dr Trade Receivables / Cr Sales. Draft: matches.
T3 · Normal	Stock bought on credit KSh 40,000. Expected: Dr Inventory / Cr Trade Payables. Draft: matches.
T4 · Rule check	Rent paid by bank KSh 25,000. Expected: Dr Rent Expense / Cr Bank. Matches only if the learner's Scene 2 rule is correct.
T5 · Rule check	Owner takes cash KSh 5,000. Expected: Dr Drawings / Cr Cash. Matches only if the Scene 2 rule is correct.
T6 · Rule check	Customer pays KSh 18,000 into the bank. Expected: Dr Bank / Cr Trade Receivables. Matches only if the Scene 2 rule is correct.
T7 · Rule check	Owner invests KSh 50,000 in cash. Expected: Dr Cash / Cr Capital. Matches only if the Scene 2 rule is correct.
T8 · Routine fault: one-sided	Supplier paid KSh 15,000 from bank. Expected: Dr Trade Payables / Cr Bank. Draft posts Cr Bank only. Fault F1: the pay-supplier routine posts one leg.
T9 · Routine fault: wrong amount	Delivery paid in cash KSh 4,500. Expected: Dr Delivery Expense 4,500 / Cr Cash 4,500. Draft debits 5,400. Fault F2: the delivery routine transposes digits.

SCENE 4 DEVELOPER SPECIFICATION (PARTS A–B, TESTING)	
T10 · Balanced but wrong	Equipment bought for cash KSh 20,000. Expected: Dr Equipment / Cr Cash. Draft posts Dr Delivery Expense / Cr Cash. Fault F3: wrong account; the trial balance would still balance.
Scoring and console	Suggested Testing & Debugging split: coverage of the four categories 40%, correct prediction before running 30%, correct revision 30%. Each mismatch offers a "which rule or routine?" picker (F1–F3 and the six Scene 2 rules), with hints after ~8s idle. Hints are recorded, not penalised.

Scene 4 developer specification (Part C, handover)

SCENE 4 DEVELOPER SPECIFICATION (PART C, HANDOVER)	
Option A · Correct	Posting rules follow account type; every posting has equal debits and credits; trial balance differences are traced through a suspense account and corrected; a balanced trial balance does not prove every entry is right.
Option B · Overclaim	If the trial balance balances, all entries are correct. Feedback: ask the learner to restate what a trial balance can and cannot show (see T10).
Option C · Incomplete	Posting rules only, with no mention of the trial balance or its limits. Prompt for the trial balance trace.
Rationale rubric (Reasoning)	One third each for naming the debit and credit rule by account type, explaining how a trial balance difference is traced, and stating what a balanced trial balance cannot show. Free text with a short template. All ten cases must pass before the handover report unlocks.

Scene [5] — [Debrief, Feedback & Closing – Can You Trust the Ledger?]

NODE [5] — [Debrief, Feedback & Closing – Can You Trust the Ledger?]	
Situation	**Visuals:** The handover is complete. The Smart Ledger workspace closes to a debrief panel showing the final score, the ten-case test report, and a timeline of the learner's own decisions, with feedback on each. Grace thanks the learner. **Audio:** brighter but restrained music; soft confirmation tones as panels open. **Grace:** "Now every entry has two sides, and the system tells me when something is off." **Narrator:** "Let us look back at your decisions. Open each card: what you chose, what happened, and what the strongest path looks like."

NODE [5] — [Debrief, Feedback & Closing – Can You Trust the Ledger?]	
Interaction	**No scored decisions.** The learner reviews the scorecard (five strand scores as numbers and labels), opens the four feedback cards (Posting Rules, Trial Balance and Errors, Testing, Explanation) to see their own choice, the result, and the expert reasoning, and reviews the test report. The learner may optionally select one lesson to carry into future work (every entry has two sides; trace a difference to its cause before clearing it; a balanced trial balance is not proof of accuracy), answer the reflection questions, and add a lesson-learned note. These are saved, not scored. The learner then selects Finish.
Response/indicator	**Feedback is specific to the learner's own path.** For example: "Your rule for rent debited Bank and credited Rent Expense. Rent is an expense, so it increases with a debit, and Bank is an asset that falls with a credit. The entry is Dr Rent Expense, Cr Bank." Common misconceptions are named when the learner's log shows them: swapping the debit and credit sides, assuming a debit always means an increase, clearing a suspense difference without tracing it, and treating a balanced trial balance as proof of accuracy. Strong paths receive confirmation and a next-level insight. **Scorecard:** the five strand scores and the final score are shown as numbers with plain-text labels. The dashboard does not label the learner as "best" or "worst." **Closing:** the narrator restates that Smart Ledger supports bookkeeping checks only, that an accountant still reviews the accounts, and that the business and figures are fictional.
Teaching content	**Consolidation only.** The debrief points back to concepts already taught in Scenes 2-4: debit and credit rules by account type, the trial balance and suspense account, error types and corrections, and testing. No new concepts are introduced, and nothing new is assessed. **Key takeaways shown after completion:** • Every transaction has two equal effects. • Debit and credit follow the account type; a debit does not always mean an increase. • A trial balance difference is a clue to trace, not a number to hide in suspense. • A balanced trial balance does not prove the books are correct. • Test the cause of an error, not just the symptom.
State change	The final score is calculated from the strand scores earned in Scenes 2-4; no points are awarded in this scene. Saves the optional reflection lesson and note and completion. Logs reflection_lesson, reflection_note,

NODE [5] — [Debrief, Feedback & Closing – Can You Trust the Ledger?]	
	scene_5_complete = true, and simulation_complete = true. The facilitator view receives the strand scores, test evidence, error patterns, and learner reflection.
Goes to	→ End of simulation → Debrief/reflection dashboard (Section 8)

Scene 5 developer specification

SCENE 5 DEVELOPER SPECIFICATION	
Final score	Weighted total (see Section 8) with a plain-text label. Suggested bands: Strong 80-100, Developing 60-79, Needs revisit below 60.
Strand scores	Five strands (Posting-Rule Accuracy, Trial Balance & Ledger Skills, Error Detection & Correction, Testing & Debugging, Reasoning), each with a numeric score and a one-line explanation of what raised or lowered it.
Feedback cards	Posting Rules, Trial Balance and Errors, Testing, Explanation. Each shows your choice, the result, why it happened, and the expert path. Choices correct the first time show confirmation plus one deeper insight.
Test report	The ten-case table with expected versus actual entries, the faults found (F1–F3), and the revisions the learner made.
Reflection	The debrief questions from Section 8, an optional lesson selection, and an optional lesson-learned note. Saved, not scored.

7 · Audio/Video Production Spec (if applicable)

PRODUCTION SPEC	
Format & duration	Interactive 2D system simulation (workspace, T-accounts, trial balance panel, and test console); 20-25 minutes total. Scene 1 (opening sequence/title card): about 1 minute, non-interactive. Scenes 2-4: about 5-7 minutes each, each with a narrated setup, a scored activity, and dynamic feedback. Scene 5 (debrief, feedback, and closing): about 3 minutes, review-only.
Voices required	Narrator: calm, clear instructional voice. Grace Achieng: a warm, practical shop owner who wants understandable books. Optional supporting voice: brief cashier or supplier lines only where needed. Use Kenyan English pronunciation consistently. Caption all dialogue.
Visual style	Clean software interface in a fictional Kenyan retail setting: rule sheet, posting table, T-accounts, trial balance panel, and test console. Realistic shop

PRODUCTION SPEC	
	details (till, receipts, supplier invoices, mobile-money cues). Restrained interface animation; no real branding or customer data.
Accessibility	Default-on subtitles for all spoken content; high contrast; clear focus states; every drag, tap, or navigation action has a keyboard/tap alternative; T-accounts and the trial balance readable as labelled tables ("debit" and "credit" announced in full); no information communicated by colour alone.
Fixed vs. open	**Fixed:** Zuri Home Supplies, Grace, the rule sheet, account list, opening balances, the month's transactions, errors E1–E2, the test bank and faults F1–F3, scoring logic, and learning outcomes. **Open:** visual layout, character appearance, sound design, and transitions, provided task logic and answer visibility are unchanged.

Full transcript

Scene 1 – Opening Sequence: Welcome to Zuri Home Supplies

NARRATOR: *Welcome to Zuri Home Supplies, a growing homeware shop in Nairobi. The shop has started using Smart Ledger, a system that posts each transaction to the accounts automatically. [AUDIO CUE: Nairobi morning ambience, door chime, till and keyboard sounds; title card: THE SMART LEDGER SYSTEM]*

GRACE: *The system records everything for me now. But the trial balance does not balance, and I cannot tell whether the postings are right. [AUDIO CUE: phone notification]*

NARRATOR: *You are joining Grace as a junior accounts systems assistant. Here is the plan: set the posting rules, trace the errors in the ledger, then test the fixes and hand over. You will be assessed on posting rules, ledger and trial balance skills, error correction, testing, and your explanation. Speed is not scored. All figures are fictional. [AUDIO CUE: music rises slightly; start prompt tone]*

NARRATOR: *Press Start when you are ready. [AUDIO CUE: title-card sting]*

Scene 2 – Set the Posting Rules

NARRATOR: *Open the rule sheet and look at what the system posted for each sample transaction. Then decide your first debugging action. [AUDIO CUE: keyboard clicks, soft interface tones]*

GRACE: *Some of these look wrong to me, but I cannot tell why. What should the system be doing? [AUDIO CUE: low investigative music bed]*

NARRATOR – WRONG FIRST ACTION: *Changing headings or hiding accounts does not show whether the postings are correct. Try again. [AUDIO CUE: neutral corrective cue]*

NARRATOR: *For each transaction type, decide which account is debited and which is credited. Think about what came in, what went out, and what kind of account each one is. [AUDIO CUE: interaction prompt tone]*

NARRATOR – REVERSED SIDES: *The accounts are right, but the sides are swapped. Rent is an expense, and expenses increase with a debit. [AUDIO CUE: soft corrective cue]*

NARRATOR – WRONG ACCOUNT: *Check what the customer's payment actually did. The shop is being paid what it was already owed, so which account falls? [AUDIO CUE: soft corrective cue]*

NARRATOR – CORRECT FEEDBACK: *Good. The debit and credit sides follow the rule for the account type, and the accounting equation stays in balance. [AUDIO CUE: soft confirmation chime]*

NARRATOR: *The posting rules are set. Now see what the system did with a whole month of transactions. [AUDIO CUE: page-turn transition]*

Scene 3 – Why the Trial Balance Fails

NARRATOR: *The corrected rules apply to new postings. But the first run left errors in the ledger. [AUDIO CUE: calculator taps, soft office ambience]*

GRACE: *The system will not let me close the month. It has parked the difference in a suspense account, but nobody knows what it is. [AUDIO CUE: low investigative music bed]*

NARRATOR: *Start by listing each account balance in the debit or credit column. [AUDIO CUE: card movement sound]*

NARRATOR – INCORRECT FEEDBACK: *Check the normal balance of this account. Assets, expenses, and drawings usually have debit balances. Liabilities, capital, and income usually have credit balances. [AUDIO CUE: neutral corrective cue]*

NARRATOR: *Debits and credits do not agree. The difference is KSh 14,100. Find out why. [AUDIO CUE: alert tone]*

NARRATOR: *Look at the accounts flagged in the ledger. Compare each one with the transaction it came from, and decide what type of error you see. [AUDIO CUE: paper comparison sound]*

NARRATOR – CORRECT FEEDBACK: *Right. One side of the supplier payment was never posted. And KSh 4,500 was keyed as 5,400, a transposition: the difference of 900 divides exactly by nine. [AUDIO CUE: positive confirmation]*

NARRATOR – INCORRECT FEEDBACK: *Compare the two sides of the entry. Is a leg missing, is a figure wrong, or are the sides swapped? [AUDIO CUE: soft corrective cue]*

NARRATOR: *Now choose a correcting entry for each error. Use the suspense account to hold what you cannot yet explain, and clear it once each error is fixed. [AUDIO CUE: interaction prompt tone]*

NARRATOR – CORRECT FEEDBACK: *Correct. Both errors are fixed, the suspense account is clear, and the trial balance agrees at KSh 465,000. [AUDIO CUE: confirmation chime]*

NARRATOR: *The ledger balances. But Grace needs to know the system will not make the same mistakes again. Time to test it. [AUDIO CUE: transition music rises]*

Scene 4 – Test, Debug and Hand Over

NARRATOR: *Choose test cases that cover typical transactions, the rules you set, the faults found in the ledger, and one case where the trial balance could balance but the entry is wrong. Predict first, then run, then compare, then revise. [AUDIO CUE: keyboard clicks, test console opens]*

GRACE: *The books balance now. How do I know the system will not make the same mistakes next month? [AUDIO CUE: soft emphasis tone]*

NARRATOR – MISMATCH: *The result differs from the rule. Which routine or rule is responsible? [AUDIO CUE: soft alert tone]*

NARRATOR – BALANCED BUT WRONG: *The trial balance still balances here. But the entry is in the wrong account. Balanced does not mean correct. [AUDIO CUE: neutral corrective cue]*

NARRATOR: *Now run all ten cases against the finished system to confirm the fixes. [AUDIO CUE: pass tones as cases succeed]*

GRACE: *If the trial balance balances, can I trust everything in the books? [AUDIO CUE: gentle emphasis tone]*

NARRATOR: *Explain the posting rules, how you traced the trial balance difference, and what a balanced trial balance cannot tell you. This system supports bookkeeping checks only; an accountant must still review the accounts. [AUDIO CUE: reflection prompt tone]*

NARRATOR: *Handover complete. Let us look back at how your decisions shaped the result. [AUDIO CUE: transition music rises]*

Scene 5 – Debrief, Feedback and Closing: Can You Trust the Ledger?

NARRATOR: *The debrief panel shows your final score, your test report, and the decisions that shaped them. [AUDIO CUE: brighter music bed]*

GRACE: *Now every entry has two sides, and the system tells me when something is off. [AUDIO CUE: music softens]*

NARRATOR: *Let us look back at your decisions. Open each card: what you chose, what happened, and what the strongest path looks like. [AUDIO CUE: dashboard navigation cue]*

NARRATOR [Posting Rules – Strong]: *You matched each transaction to the correct debit and credit by account type. [AUDIO CUE: soft feedback tone]*

NARRATOR [Posting Rules – Developing]: *Most rules were correct, but some needed a second look at the account type or the direction of the entry. [AUDIO CUE: soft feedback tone]*

NARRATOR [Posting Rules – Needs revisit]: *Several rules put the debit or credit on the wrong side. Ask what came in, what went out, and what type of account each one is. [AUDIO CUE: soft feedback tone]*

NARRATOR [Trial Balance – Strong]: *You placed every balance in the right column and read the difference correctly. [AUDIO CUE: soft feedback tone]*

NARRATOR [Trial Balance – Developing]: *You placed most balances correctly, but the size or direction of the difference needed another look. [AUDIO CUE: soft feedback tone]*

NARRATOR [Trial Balance – Needs revisit]: *Several balances went in the wrong column. Check each account's normal balance before totalling. [AUDIO CUE: soft feedback tone]*

NARRATOR [Error Correction – Strong]: *You identified each error type and chose correcting entries that cleared the suspense account. [AUDIO CUE: soft feedback tone]*

NARRATOR [Error Correction – Developing]: *You traced most errors, but a correcting entry left part of the suspense balance uncleared. [AUDIO CUE: soft feedback tone]*

NARRATOR [Error Correction – Needs revisit]: *Errors were not traced to their cause before correcting. Compare both sides of each entry with its source transaction. [AUDIO CUE: soft feedback tone]*

NARRATOR [Testing – Strong]: *You covered all four test categories, predicted before running, and revised the right routine each time. [AUDIO CUE: soft feedback tone]*

NARRATOR [Testing – Developing]: *Your tests covered most categories, but some predictions came after running or a revision targeted the wrong rule. [AUDIO CUE: soft feedback tone]*

NARRATOR [Testing – Needs revisit]: *Important categories were left untested. Include faults found earlier and cases where the trial balance could still balance. [AUDIO CUE: soft feedback tone]*

NARRATOR [Explanation – Strong]: *Your handover named the rules, the trial balance trace, and the limits of a balanced trial balance. [AUDIO CUE: soft feedback tone]*

NARRATOR [Explanation – Developing]: *Your explanation covered the rules, but the trace or the limits of a trial balance were thin. [AUDIO CUE: soft feedback tone]*

NARRATOR [Explanation – Needs revisit]: *Your explanation overclaimed or left out key points. A balanced trial balance does not prove every entry is right. [AUDIO CUE: soft feedback tone]*

NARRATOR: *If you wish, select the lesson you would carry into future work: every entry has two sides; trace a difference to its cause before clearing it; or a balanced trial balance is not proof of accuracy. You can add a note too. It is saved but not scored. [AUDIO CUE: reflection prompt tone]*

GRACE: *Thank you for staying with the numbers until they made sense. [AUDIO CUE: closing music begins to fade]*

NARRATOR: *Simulation complete. Smart Ledger supports bookkeeping checks only, and an accountant must still review the accounts. The business and figures in this simulation are fictional. [AUDIO CUE: closing music fade out]*

8 · Debrief & Leaderboard

How tracked strands combine into a final score

Final score = (Posting-Rule Accuracy x 0.30) + (Trial Balance & Ledger Skills x 0.15) + (Error Detection & Correction x 0.25) + (Testing & Debugging x 0.25) + (Reasoning x 0.05).

Worked example

PR 80, LT 90, ED 75, TD 85, RS 80 gives 24 + 13.5 + 18.75 + 21.25 + 4 = 81.5 / 100.

Scoring interpretation

The dashboard displays strand scores and the overall score numerically with plain-text labels (Strong 80-100, Developing 60-79, Needs revisit below 60). It does not label the learner as "best" or "worst." Feedback focuses on the specific skill evidenced by each score.

What's displayed but not scored

Test cases attempted and passed, categories covered, hints and retries, time spent, and the optional reflection lesson and note captured in Scene 5. Do not score speed.

Backend (Supabase / Firebase)

Recommended. Suggested fields: learner_id, simulation_id, timestamps, scene_reached, audit_action_choice, rule_mapping_choices, trial_balance_columns, difference_identified, error_types_selected, correcting_entries, suspense_cleared, cases_chosen, predictions, actual_results, rule_revisions, handover_summary_choice, rationale, hints_used, retries, strand_PR, strand_LT, strand_ED, strand_TD, strand_RS, final_score, reflection_lesson, reflection_note, completed_at. Synthetic data only.

Facilitator dashboard needs

Show cohort averages by strand, common error categories (swapped sides, wrong account, clearing suspense without tracing, treating a balanced trial balance as proof of accuracy), and item-level attempt patterns without exposing learner personal data unnecessarily.

Debrief questions

1. Which transaction type was easiest to get the debit and credit sides wrong on, and why?
2. How did you decide whether the trial balance difference came from one error or more than one?
3. Why does clearing a suspense account not fix the cause of an error?
4. Which test case showed an error that a balanced trial balance would not reveal?
5. When would you trust a system's automatic posting, and what checks would you keep?

9 · Accessibility Requirements

☒ Default-on subtitles for all voiced content, minimum 18px, high contrast
☒ Keyboard or tap/click alternative for every drag/hold gesture
☒ Text-label equivalents for directional/compass navigation
☒ Accessible labelling for timers and data-entry fields
☒ Minimum 44px touch targets throughout
☒ Do not rely on colour alone to distinguish correct, incorrect, or pending states

☒ Provide a visible calculator-friendly input method and allow keyboard entry for all accounting amounts

☒ Provide text equivalents for all audio cues that carry task-relevant information

☒ Keep feedback concise and readable; never hide corrective guidance only in animation

☒ Scene 5 debrief panels must be fully keyboard-navigable, and every scorecard value and snapshot item must show a text value and label, not colour alone

☒ The Scene 5 reflection note should support typing as well as preset lesson choices

☒ Provide a non-drag alternative for Scene 2 rule mapping and Scene 3 trial balance sorting (select an account, then choose Debit or Credit)

☒ T-accounts and the trial balance must be readable as labelled tables, with “debit” and “credit” announced in full by screen readers